



7th Global AI Hackathon

In collaboration with MIT Club of Northern California and MIT Club of Germany

C H A L L E N G E

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10 × FASTER SCIENTIFIC DISCOVERY

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# Agentic Scientific Discovery

*Build an AI lab for the next Nobel Prize caliber breakthrough*

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POWERED BY



**Build with Omnigent**

Required for every challenge submission · 24 hour scientific discovery challenge

## Goals and Motivation

*Build an agentic AI lab for scientific breakthroughs*

### REQUIRED PLATFORM

**Omnigent** (managed on Databricks or open source)

AI has already helped unlock Nobel Prize-winning science. By 2024, AlphaFold2 had enabled predictions for almost 200 million protein structures; its creators shared the Nobel Prize in Chemistry. That scale of discovery is the inspiration for this challenge.

### THE MOONSHOT

**Build the Omnigent AI lab behind the next Nobel Prize-caliber breakthrough and help make scientific discovery 10× faster.**

In the next 24 hours, build a working scientific lab around one problem you care deeply about. Show how coordinated agents generate insight, formulate hypotheses, design experiments, run computational tests or analyses, interpret results and decide what to investigate next.

Scientific inspiration: [Royal Swedish Academy of Sciences, 2024 Nobel Prize in Chemistry](#)

## Choose Any Science Domain You Are Passionate About

Your project can address any area of science: biology, medicine, materials, energy, climate, agriculture, chemistry, physics, AI research, neuroscience, astronomy or something entirely different. Choose one specific scientific question your prototype can meaningfully investigate within 24 hours.

Choose a question that can be tested with existing scientific datasets, published literature, APIs, simulations, models, computational experiments, benchmark tasks or data you generate yourself. If you can generate new data during the hackathon, you are warmly invited to do so - but it is not required.

A physical or wet-lab experiment may be the ultimate next step, but teams are not expected to complete wet-lab validation during the hackathon.

### EXAMPLES

Discover a promising therapeutic mechanism, identify a new material, explore an energy system, test an AI research idea, investigate a climate or agriculture question, or bring a scientific frontier of your own.

### FOCUS

Focus on a question where your prototype can produce evidence, learn from it and make a better next scientific decision.

## Orchestrate Your Scientific Lab with Omnigent

*Mandatory for every submission: build with Omnigent*

Omnigent must orchestrate the live discovery workflow. Demonstrate multiple specialist agents exchanging outputs, using tools and adapting their plan after an experimental result.

### AI builders — how to get started with Omnigent

#### CHOOSE YOUR SETUP

Use managed Databricks or open-source Omnigent from GitHub. A Databricks account is only required for the managed Databricks route.

#### LAUNCH OMNIGENT

**Databricks:** sign in → open <workspace-url>/omnigent → New session → Sandbox. **GitHub:** install Omnigent → run omnigent → select your model or agent harness.

#### BUILD + DEMO THE DISCOVERY LOOP

Connect your data and tools, define specialist-agent handoffs, run an experiment, and show how the result changes what the agents investigate next.

#### GET STARTED

[Databricks login](#) · [Managed quickstart](#) · [Open-source GitHub](#)



Safety agent flags risks and routes decisions for human approval. Cyan arrows: discovery loop. Gray connections: Omnigent coordination. Adapt these suggested agent roles to your field.

## Design the handoffs that accelerate discovery

For each agent, specify the scientific decision it owns, the tools it can use, its inputs and its output. Pass structured evidence, candidate IDs, experiment specifications and results between agents. Keep a shared research record so every decision can be reconstructed.

Run independent searches or experiments in parallel. Let surprising results reopen an earlier assumption. Give the planner a budget and make it choose between competing tests. Build the workflow your science needs, and add or combine roles where that improves the experiment.

### HUMAN APPROVAL

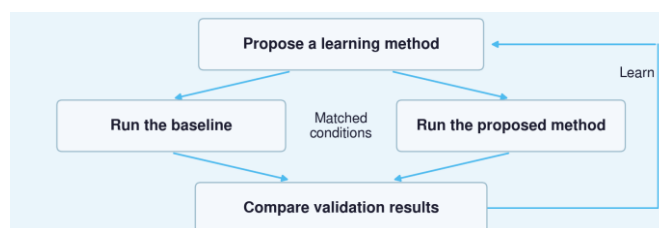
Scientists set the objective and approve consequential actions. A safety agent can flag risks and request approval; enforce the boundary through tool permissions and Omnigent policies.

## Run an Experiment That Changes the Next Decision

Start with a clear question and a measurable scientific outcome. Use literature, structured data, simulations, models or generated data to develop a hypothesis. Design at least two possible tests and choose one using expected learning, feasibility and cost. Run the test, interpret the result and use what you learned to determine the next step.

### What counts as an experiment in this challenge

Your experiment can be a simulation, computational screening experiment, benchmark evaluation, model comparison, analysis of an existing experimental dataset, sensitivity or counterfactual analysis, experiment on data you generate yourself, or another reproducible computational test. The important requirement is that it produces evidence that changes the next scientific decision.



A physical or wet-lab experiment may be proposed as the next step, but it is not required. Adapt this experimental logic to the scientific frontier you choose.

## Connect the sources your science needs

|                                   |                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ACROSS DISCIPLINES</b>         | OpenAlex helps agents connect publications, citations and research communities.                                                                |
| <b>DRUG DISCOVERY AND BIOLOGY</b> | <a href="#">Europe PMC</a> supplies biomedical literature; <a href="#">PubChem</a> supplies compound identities, properties and assay records. |
| <b>MATERIALS AND ENERGY</b>       | <a href="#">Materials Project</a> and NIST JARVIS provide structures and computed properties to support candidate selection and simulation.    |
| <b>AI RESEARCH</b>                | <a href="#">arXiv</a> provides research papers; OpenML provides datasets and benchmark tasks for reproducible experiments.                     |

These are starting points, not restrictions. Use the scientific infrastructure your problem requires. If you can generate useful data through simulation, synthetic experimentation, model outputs or real measurements during the hackathon, you are encouraged to do so. Check access keys, licenses and compute needs early.

## Show Progress Toward 10× Faster Discovery

The long-term moonshot is to make scientific discovery 10× faster. You do not need to prove a 10× improvement within 24 hours. Instead, identify one bottleneck in scientific discovery and demonstrate how your agentic lab helps compress, automate or improve it.

### Define what faster discovery means for your problem

Examples include screening more candidates within the same time or compute budget; reducing downstream tests; shortening the path from literature to a testable hypothesis; evaluating more hypotheses in parallel; reducing human effort; or improving how quickly new evidence changes the next decision.

**Report the improvement you actually observe.** It may be 1.5×, 3×, 5× or 10×. The strength of your evidence matters more than claiming the largest multiplier.

#### ONE COMPLETE DISCOVERY LOOP

**Question → Evidence → Hypothesis → Experiment → Result → Updated decision**

Explain what bottleneck you attacked, what the agents learned, what should happen next and what would need to happen for the workflow to approach 10× acceleration at scale.

### Build toward one convincing demonstration

#### First 4 hours

Choose the scientific domain, research question, measurable outcome, data and tools, and the discovery bottleneck you want to improve. Make sure you can access the data, run a meaningful test in 24 hours, measure the result and use it to change the next decision.

#### Next 14 hours

Build the Omnigent workflow, generate hypotheses or candidates and select experiments. Get one complete discovery loop running, including a result that changes the next decision.

#### Final 6 hours

Strengthen the experiment, analyze the result and prepare a two minute demo. Show the question, agent handoffs, experiment, result, what the lab learned and what it would investigate next.

## What Success Looks Like

#### REQUIRED: BUILD WITH OMNIGENT

Show purposeful collaboration between specialist agents.

#### SCIENTIFIC PROGRESS

An ambitious question and a meaningful, reproducible result.

#### DISCOVERY ACCELERATION AND LEARNING

Identify a meaningful bottleneck, demonstrate measurable progress and justify the next experiment from what the lab learned.

Require citations for factual claims. Attach source evidence or run records, label agent-generated hypotheses, preserve uncertainty, document controls and human approval gates, and state the validation still needed before real-world use.

## Evaluation and Submission

**30%**

Omnigent orchestration

**25%**

breakthrough potential

**20%**discovery acceleration  
and learning**15%**

scientific rigor

**10%**creativity and  
responsibility

#### SUBMIT

The repository, agent specifications and policies, a two minute demo, cited evidence, experiment code and results, your measured improvement and your next experiment.

*Our world needs breakthroughs. In the next 24 hours, build the scientific lab that could help make the next one possible. Pick the science you care about. Build the agents. Run the experiment. Learn from the result. Show us the path to 10× faster discovery.*